

Classical Mechanics 1, Autumn 2025 CMI

Assignment 3

Due by 6pm, Fri 29 Aug, 2025

trajectory, gradient

1. **⟨1 + 4⟩** Suppose a particle has a trajectory in 3d space given by the position vector $\mathbf{r}(t) = \cos t \hat{x} + \sin t \hat{y} + t \hat{z}$. (a) Find the velocity and acceleration vectors. (b) Plot the trajectory and describe its shape in words. Indicate the directions of the velocity and acceleration vectors at a point on the trajectory and say in words which way they point.
2. **⟨4 + 5⟩ Gradient of a scalar field.** Consider the scalar field $\psi = 1/\sqrt{x^2 + y^2 + z^2}$ defined using Cartesian coordinates in $\mathbb{R}^3$ with the origin left out. (a) Find the gradient of ψ for x, y, z not all zero. After obtaining the answer in Cartesian coordinates, re-express ψ and the final answer for the gradient in spherical polar coordinates. (b) Say in words which way $\nabla\psi$ points and why and how its magnitude varies. Plot this vector field on a suitably chosen plane through the origin. Does the choice of plane matter? Why or why not?
3. **⟨2 + 2 + 4⟩ Force from a potential.** Suppose a potential in 3d Euclidean space is given by $V = \alpha/r^n$ where r is the radial distance from the origin, $\alpha > 0$ a real constant and n a real number. (a) Find the associated force $\mathbf{F} = -\nabla V$ and express it in spherical polar coordinates. (b) For $n = -2$ write V and $\mathbf{F}$ in Cartesian coordinates. (c) Describe the resulting $\mathbf{F}$ for $n = -2$ by saying whether the force is attractive/repulsive (relative to the origin), how it behaves with distance and give it a suitable name using standard terminology. Plot the graph of $V(r)$.